

Selection indexes in plant breeding

Métodos de melhoramento LGN-5801

José Tiago Barroso Chagas

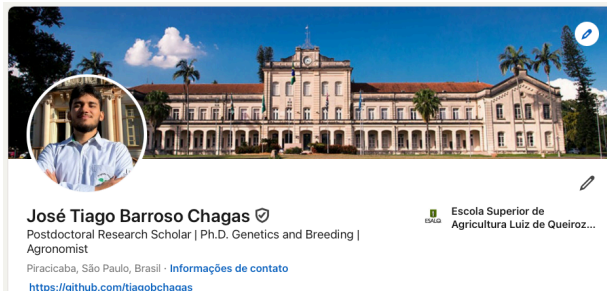
Escola Superior de Agricultura Luiz de Queiroz - Universidade de São Paulo

2025-11-18

Outline I

- 1 Experimental Design
- 2 Mixed models equation
- 3 Single and Multienvironment trials
- 4 Selection indexes
- 5 Bayesian probabilistic selection index (BPSI)

Background



*Agronomist (**UFCA**- 2018)*

*Msc. in Genetics and plant breeding (**UENF**-2020)*

*Dsc. in Genetics and breeding (**UFV**- 2024)*

*Postdoctoral research scholar (**UFV**-2024-2025); (**ESALQ/USP**-2025)*

Section 1

Experimental Design

Main principles of experimental design

- Replication
- Randomization
- Local control

Completely Randomized Design 3X5

ROWS	3	T5	T5	T5	T4	T3
	2	T1	T1	T1	T2	T3
	1	T2	T2	T4	T4	T3
		1	2	3	4	5
		COLUMNS				

Randomized Complete Block Design 3X5

ROWS	3	T3	T5	T2	T4	T1
	2	T5	T4	T3	T1	T2
	1	T2	T4	T1	T3	T5
		1	2	3	4	5
		COLUMNS				

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ROWS

COLUMNS

Randomized Complete Block Design 3X5

3	T3	T5	T2	T4	T1
2	T5	T4	T3	T1	T2
1	T2	T4	T1	T3	T5
	1	2	3	4	5

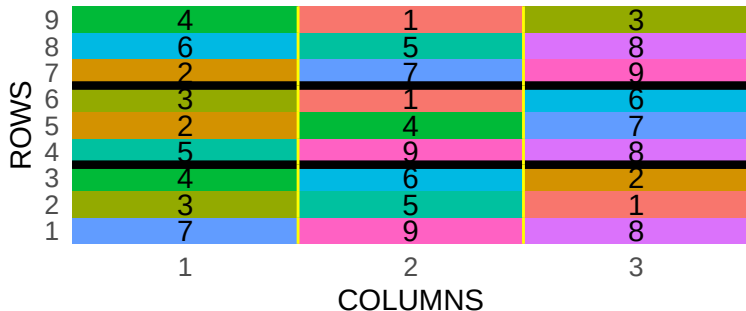
ROWS

COLUMNS

Square lattice

- Many test genotypes to evaluate in RCBD
- k treatments by k blocks
- (9treat/k3/rep3)

Square Lattice Design Field Layout 9X3



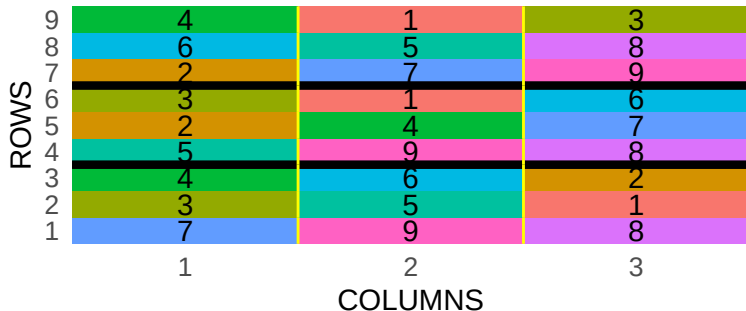
	ID	LOCATION	PLOT	REP	IBLOCK	UNIT	ENTRY	TREATMENT
1	1	PIRACICABA	1001	1	1	1	7	G-7
2	2	PIRACICABA	1002	1	1	2	3	G-3
3	3	PIRACICABA	1003	1	1	3	4	G-4

ESALQ

Square lattice

- Many test genotypes to evaluate in RCBD
- k treatments by k blocks
- (9treat/k3/rep3)

Square Lattice Design Field Layout 9X3



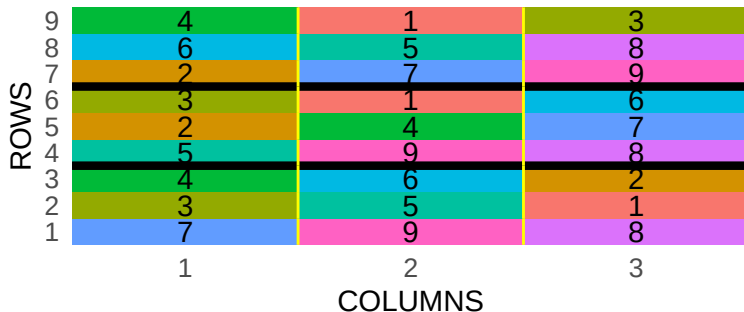
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Augmented randomized complete block design

- Trials with few seeds available
- Large number of treatments
- (30gen/3check/k6)

	ID	LOCATION	PLOT	ROW	COLUMN	CHECKS	BLOCK	ENTRY
1	1	PIRACICABA	1001	1	1	1	1	3
2	2	PIRACICABA	1002	1	2	1	1	2
3	3	PIRACICABA	1003	1	3	0	1	23

Augmented RCBD Layout 6 x 8

6	31	2	1	32	22	11	27	3
5	16	28	5	2	10	14	1	3
4	30	1	20	2	3	19	7	29
3	13	6	1	9	3	17	25	2
2	2	3	33	26	1	12	24	15
1	3	2	23	8	1	18	4	21
	1	2	3	4	5	6	7	8

ALQ

Augmented randomized complete block design

- Trials with few seeds available
- Large number of treatments
- (30gen/3check/k6)

	ID	LOCATION	PLOT	ROW	COLUMN	CHECKS	BLOCK	ENTRY
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Augmented RCBD Layout 6 x 8

ROWS	6	31	2	1	32	22	11	27	3
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		COLUMNS							

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		COLUMNS							

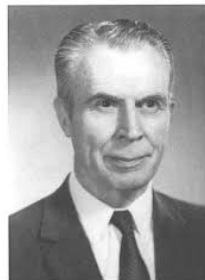
Section 2

Mixed models equation

Mixed models equation

Henderson (1953) (animal breeder) propose the mixed models equations.

Enables Best linear unbiased prediction (BLUP) and Best Linear Unbiased Estimator (BLUE).



C.R. Henderson

Mixed models equation

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}u + \mathbf{e}$$

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{X} & \mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K}^{-1} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{b}} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{Z}'\mathbf{y} \end{bmatrix}$$

Restricted maximum likelihood - (REML)

Patterson and Thompson (1971) Restricted Maximum Likelihood (REML) of random u and fixed effects $\boldsymbol{\beta}$.

Maximum likelihood function

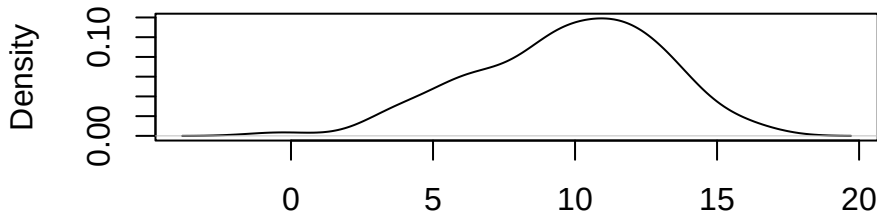
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

The maximum likelihood function seeks to optimize the values of the mean μ and variance σ , given the observations and considering normal distribution.

Mixed models equation

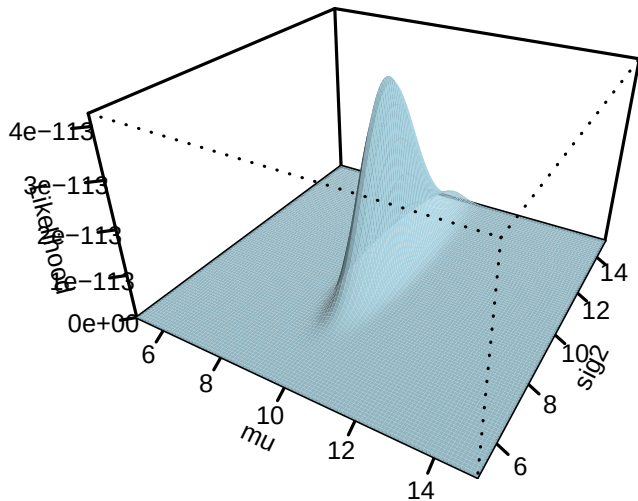
Neste exemplo consideramos 100 valores aleatórios gerados de uma distribuição normal com média 10 (μ) e desvio padrão $\sqrt{10}$ (σ)

Normal density



N = 100 Bandwidth = 1.159

Maximum likelihood function



Section 3

Single and Multienvironment trials

Simulating phenotypic traits



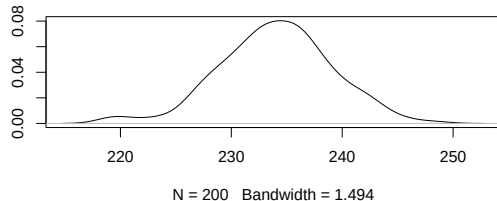
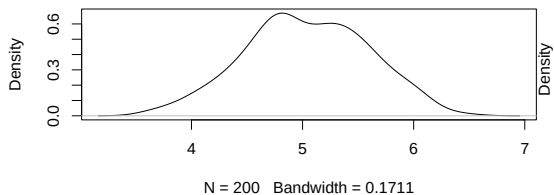
Figure 1: Grain yield (kg ha^{-1}) of maize hybrids in simulated trials



Figure 2: Plant Height (cm) of maize hybrids in simulated trials

Simulating phenotypic traits

2 traits; 100 genotypes; 2 blocks



Solving mixed models equation

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{X} & \mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K}^{-1} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{Z}'\mathbf{y} \end{bmatrix}$$

Table 1: mixed models matrices

(a) X	
block1	block2
1	0
1	0
1	0

(b) Z		
id1	id2	id3
0	0	0
0	0	0
0	0	0

(c) y
pheno
5.513532

(d) lambda	
geno	erro
0.0948	0.2075



Solving mixed models equation

Table 2: algebra with mixed models matrices

(a) X_pX

	block1	block2
block1	100	0
block2	0	100

(b) X_pZ

	id1	id2
block1	1	1
block2	1	1

(c) Z_pX

	block1	block2
id1	1	1
id2	1	1

(d) Z_pZ

	id1	id2
id1	2	0
id2	0	2

(e) $X_p y$

block1	498.0205
block2	509.2963

(f) $Z_p y$

	x
	8.834306
	10.478144

Solving mixed models equation

Table 3: LHS matrice

	block1	block2	id1
block1	100	0	1.000000
block2	0	100	1.000000
id1	1	1	4.188819

Table 4: RHS matrice

	[,1]
block1	498.020468
block2	509.296324
id1	8.834306
id2	10.478144

Table 5: sol matrice

	[,1]
block1	4.98020468
block2	5.09296324
id1	-0.29575450
id2	0.09668033

Solving mixed models equation (LME4)

Solving mixed models with package LME4

Table 6: RHS

x
-0.2957545
0.0966803
0.3478739
-0.0073687
-0.0808127
-0.2544430

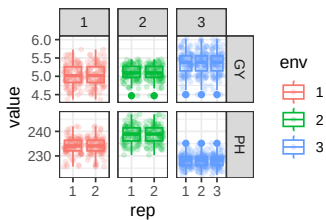
[1] 4.980205 5.092963

Table 7: LME4

	x
id.(Intercept)1	-0.3126610
id.(Intercept)2	0.1022069
id.(Intercept)3	0.3677597
id.(Intercept)4	-0.0077899
id.(Intercept)5	-0.0854323
id.(Intercept)6	-0.2689879
(Intercept)	4.9802047
rep2	0.1127586

Multienvironmental trials

Table 8: GY and PH evaluated in multienvironmental trials



$$\mathbf{y} = \mathbf{X}_1\mathbf{e} + \mathbf{X}_2\mathbf{be} + \mathbf{Z}_1\mathbf{g} + \mathbf{Z}_2\mathbf{ge} +$$

$[\mathbf{g} \sim N(0, \sigma_g^2 \mathbf{I}_n)]$ and $[\mathbf{ge} \sim N(0, \sigma_{ge}^2 \mathbf{I}_{np})]$, where n is the number of genotypes and p is the number of environments.

Multienvironment trials

2 traits; 100 genotypes; 2-3 blocks; 3 environments

Testing genotype and genotype $\times$ environment interaction effects along traits.

$$LRT = -2(\text{Log}L - \text{Log}L_r)$$

where $\text{Log}L$ é o ponto de máxima da função de verossimilhança residual do modelo completo e $\text{Log}L_r$ é o mesmo para o modelo reduzido.

Multienvironment trials

```
[1] 43.31988
```

```
Data: pheno_df
```

```
Models:
```

```
mmcr: y.GY ~ env + rep:env + (1 | id)
```

```
mmc: y.GY ~ env + rep:env + (1 | id) + (1 | id:env)
```

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
mmcr	9	1096.1	1137.0	-539.03	1078.1			
mmc	10	1054.3	1099.8	-517.15	1034.3	43.758	1	3.717e-11 ***

```
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```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Multienvironment trials

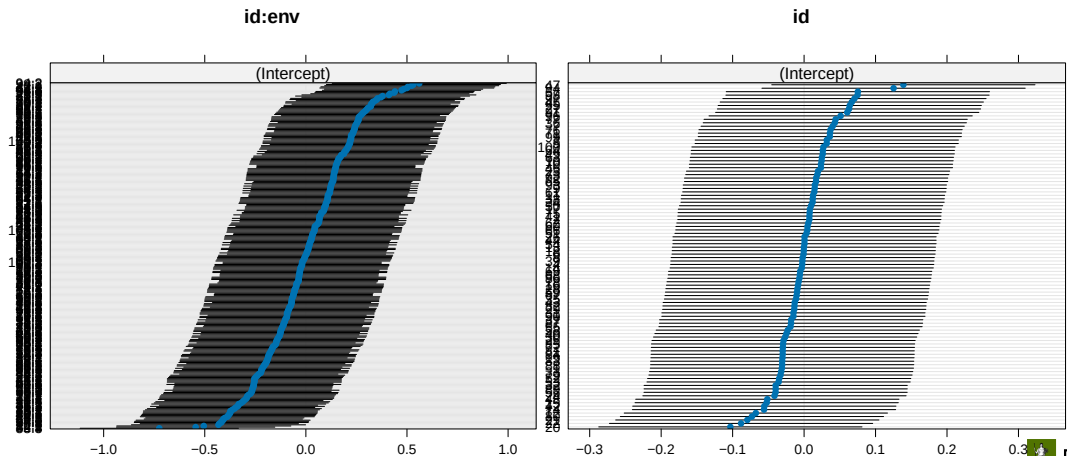
Table 9: LME4 - g x e

	(Intercept)
id:env.1:1	-0.2994339
id:env.1:2	-0.0883629
id:env.1:3	0.1092676
id:env.2:1	0.0992936
id:env.2:2	-0.0491773
id:env.2:3	0.0160696

Table 10: LME4 - g

	(Intercept)
id.1	-0.0310489
id.2	0.0073780
id.3	0.0317793
id.4	-0.0026264
id.5	0.0316266
id.6	-0.0007239

Multienvrionmental trials



Section 4

Selection indexes

Selection indexes

Indexes

Several indexes were proposed with and without restrictions (Smith (1936) & Hazel (1943) ; Kempthorne and Nordskog (1959)). We take a deep look in Factor analysis and ideotype design index (FAI-BLUP)(Rocha, Machado, and Carneiro 2018) and Bayesian probabilistic selection index (BPSI)(Chagas et al. 2025).

Indexes

Smith e Hazel

$$\underline{\hat{b}} = \hat{P}^{-1} \hat{G} \hat{a};$$

where $\hat{P}$ and $\hat{G}$ are phenotypic and genotypic variance-covariance matrix, respectively, and $\hat{a}$ are relative economic values.

Desire

$$\underline{b} = G^{-1} \underline{q}$$

where G is the genotypic variance-covariance matrix and q is the vector of desired gains (Pešek and Baker 1969)

Desire and SH

Table 11: smith-hazel and desire indexes

(a) ebvMat			(b) Var_Covar_ebv				(c) economic weights
	PH_1	PH_2					x
			11.46	6.34	0.90	-0.08	-1
			6.34	12.75	-0.35	0.28	-1
39	238.25	239.84	0.90	-0.35	5.88	0.08	-1
4	226.52	232.95	-0.08	0.28	0.08	0.09	2
							2
							2

Desire and SH

Table 12: results smith-hazel and desire

(a) economic weights

0.3030121
-0.6317332
-0.7779787
6.9098122
15.9867062
3.9098938

(b) optimum weights

0.2303050
-0.6263726
-0.8032517
7.2542133
16.1879586
5.1908039

Desire and SH

Table 13: weights smith-hazel and desire

(a) gain SH			(c) gain DS		
	x			x	
PH_1	0.3386552	(b) SH	PH_1	0.2606474	(d) DS
PH_2	0.5247615		PH_2	0.5924429	
PH_3	-0.6109782		PH_3	-0.6445062	
GY_1	0.3010492		GY_1	0.2488413	
GY_2	0.2870106		GY_2	0.2344179	
GY_3	0.2522658		GY_3	0.2209354	
		x			x
		1.262474			1.380423

Factor analysis

Maize simulated dataset (2 traits * 3 enviroments)

To proceed factor analysis, we should do the pearson correlation between traits and create matrix correlation.

	GY_1	GY_2	GY_3	PH_1	PH_2	PH_3
Blups per trait:env						
1	-0.30	-0.09	0.11	-0.80	-3.24	1.54
2	0.10	-0.05	0.02	-2.69	-2.24	1.39
3	0.35	0.04	-0.11	0.70	-0.42	1.21
	GY_1	GY_2	GY_3	PH_1	PH_2	PH_3
Blups per trait:env standarized						
1	-1.38	-0.40	0.50	-0.42	-1.53	0.81
2	0.46	-0.22	0.07	-1.43	-1.06	0.73
3	1.63	0.19	-0.51	0.37	-0.20	0.63

Análise de Fatores

	GY_1	GY_2	GY_3	PH_1	PH_2	PH_3
GY_1	1.00	-0.05	-0.15	0.10	0.15	-0.03
GY_2	-0.05	1.00	0.04	-0.09	0.23	0.15
GY_3	-0.15	0.04	1.00	0.09	-0.25	0.39
PH_1	0.10	-0.09	0.09	1.00	0.13	-0.26
PH_2	0.15	0.23	-0.25	0.13	1.00	-0.34
PH_3	-0.03	0.15	0.39	-0.26	-0.34	1.00

Correlation near to zero leading to different factors. Representing correlation among traits
 Correlações próximas de zero propiciam a extração de fatores diferentes.
 Representar as correlações entre os traits por meio de variáveis latentes (fatores).

Análise de Fatores - Autovalores

Determinação dos autovalores pela seguinte equação:

$$[\det(\lambda^2.I - \rho) = 0]$$

Multiplicado por uma matriz identidade diminuida da matriz de correlações ρ

$$\begin{bmatrix} \lambda^2 & 0 & 0 & \cdots & 0 \\ 0 & \lambda^2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda^2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda^2 \end{bmatrix} \begin{bmatrix} \rho_{11} & \rho_{12} & \rho_{13} & \cdots & \rho_{1n} \\ \rho_{21} & \rho_{22} & \rho_{23} & \cdots & \rho_{2n} \\ \rho_{31} & \rho_{32} & \rho_{33} & \cdots & \rho_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho_{n1} & \rho_{n2} & \rho_{n3} & \cdots & \rho_{nn} \end{bmatrix} \cdots \begin{bmatrix} \lambda^2 - \rho_{11} & -\rho_{12} & -\rho_{13} & \cdots \\ -\rho_{21} & \lambda^2 - \rho_{22} & -\rho_{23} & \cdots \\ -\rho_{31} & -\rho_{32} & \lambda^2 - \rho_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \\ -\rho_{n1} & -\rho_{n2} & -\rho_{n3} & \cdots \end{bmatrix}$$

Análise de Fatores - Polinômio característico

Obtenção do polinômio característica de ordem relativa à quantidade de fatores.

(n características - ordem n)

Error term: 1.070302

Polinômio característico

$0.5352 - 3.9883x + 11.8003x^2 - 17.7752x^3 + 14.4279x^4 - 6x^5 + x^6$

Raízes do polinômio - Autovalores

[1] 1.7653 1.2008 1.0624 0.9772 0.5603 0.4340

Os *auto valores* são as raízes do polinômio característico da matriz de correlações.

A soma dos autovalores é igual a quantidade de variáveis originais

[1] 6

Análise de Fatores

Os autovetores são derivados dos autovalores e resultam de um sistema de equações.

$$[det(\lambda^2.I - \rho)\mathbf{v}_i = 0]$$

$$\begin{bmatrix} \lambda^2 - \rho_{11} & -\rho_{12} & -\rho_{13} & \cdots & -\rho_{1n} \\ -\rho_{21} & \lambda^2 - \rho_{22} & -\rho_{23} & \cdots & -\rho_{2n} \\ -\rho_{31} & -\rho_{32} & \lambda^2 - \rho_{33} & \cdots & -\rho_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -\rho_{n1} & -\rho_{n2} & -\rho_{n3} & \cdots & \lambda^2 - \rho_{nn} \end{bmatrix} \begin{bmatrix} v_{i1} \\ v_{i2} \\ v_{i3} \\ \vdots \\ v_{in} \end{bmatrix} = 0$$

$$(\lambda^2 - \rho_{11})v_{i1} - \rho_{12}v_{i2} - \rho_{13}v_{i3} - \cdots - \rho_{1n}v_{in} = 0]$$

Análise de Fatores - Escores fatoriais

Obtido por meio do *autovalor* dividido pela raiz quadrada do seu respectivo *autovetor*.

Indica relação com as variáveis originais.

$$S_{11} = \frac{v_{11}}{\sqrt{\lambda_1^2}}$$

Extração dos fatores

Os escores fatoriais multiplicados pelas variáveis padronizadas (Zscore) geram os fatores.

A grandeza dos escores para cada variável representará o seu peso em cada fator.

$$S_{11} = \frac{v_{11}}{\sqrt{\lambda_1^2}}$$

Análise de Fatores- Cargas fatoriais

Representam as correlações de pearson entre os fatores extraídos e as variáveis originais (+carga = +importância da variável em cada fator).

O resultado da soma das cargas fatoriais ao quadrado de cada fator é próprio autovalor do qual ele foi extraído.

Mais informações sobre a história da análise fatorial nesse artigo Bartholomew (1995)

Índice de Seleção

$$P_{ij} = \frac{\frac{1}{d_{ij}}}{\sum_{i=1; j=1}^{i=n; j=m} \frac{1}{d_{ij}}}$$

P_{ij} probabilidade do i-ésimo genótipo ser similar ao j-ésimo ideótipo, d_{ij} distância genótipo-ideótipo euclidiana padronizada.

Section 5

Bayesian probabilistic selection index (BPSI)

Hypothesis

Hypothesis

- Probability of superior performance (Dias *et al.*, 2022)
- Selection Index (Classify progenies)

Goals

- i) extend the probabilistic method to a selection index (BPSI)
- ii) select families from carioca common bean recurrent selection to extract pure lines using BPSI index.

Bayesian Probabilistic selection index

BPSI index uses the probability of superior performance to estimate the chance of a genotype being selected in multienvironmental trials Dias et al. (2022).

$$Pr(\hat{g}_i \in \Omega|y) = \frac{1}{s} \sum_{s=1}^s I(\hat{g}_i^{(s)} \in \Omega|y) \quad (1)$$

where $\hat{g}_i$ is the genotypic value, Ω is a subset of genotypes with superior performance and s represents each sample of posterior distribution.

Bayesian Probabilistic selection index

$$BPSI_i = \sum_{m=1}^t RankProSup_m$$

where i represents the rank from probability of superior performance for each genotype evaluated and m the amount of traits.

Table 14: Hypothetical scheme of three genotypes evaluated for three traits

Genotype	T1(Rank)	T2(Rank)	T3(Rank)	PSI
1	10	5	2	i. = 17
2	5	3	10	i. = 18
3	7	3	10	i. = 19

Bayesian Probabilistic selection index

Probability of superior performance within environments

$$Pr(\widehat{g}_{ij} \in \Omega_j | y) = \frac{1}{S} \sum_{s=1}^S I(\widehat{g}_{ij}^{(s)} \in \Omega_j | y)$$

Pairwise probability of superior performance

$$Pr(\widehat{g}_i > \widehat{g}_{i'} | y) = \frac{1}{S} \sum_{s=1}^S I(\widehat{g}_i^{(s)} > \widehat{g}_{i'}^{(s)} | y)$$

Plant material and experimental design

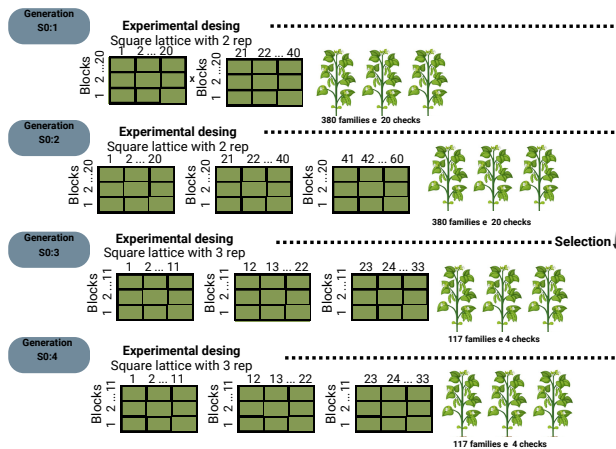


Figure 3: Experimental design

Traits evaluated



Figure 4: Grain Appearance

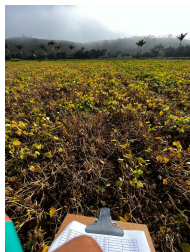


Figure 5: Plant Architecture



Figure 6: Grain Yield

First step: Adjusted means

Individual analysis was realized in each trial to get genotypes adjusted means:

$$\mathbf{Y} = \mathbf{X}_1\mathbf{r} + \mathbf{X}_2\mathbf{g} + \mathbf{Z}\mathbf{b} + \epsilon$$

where $\mathbf{Y}_{(n \times 1)}$ is vector of observations, $\mathbf{X}_{1(n \times r)}$ is a incidence matrix $\mathbf{r}$ of replication effect assumed as fixed, $\mathbf{X}_{2(n \times rk)}$ is a incidence matrix $\mathbf{g}$ of genotype effect assumed as fixed, $\mathbf{Z}_{(n \times k)}$ is a incidence matrix $\mathbf{k}$ of block effect assumed as random and $\epsilon_{(n \times 1)}$ is a vector of random errors.

The variance probability density related to model terms are: $\mathbf{b} \sim N(0, \sigma_b^2 \mathbf{I}_k)$

Second step: Bayesian model

The probabilistic model proposed by Dias et al. (2022) was fitted to each trait and setting priors to variance components of genotype, environment, and $(G \times E)$ interaction.

$$E[Y_{ij}] = \mu + g_i + a_j + (ga)_{ij} + \epsilon_{ij}$$

where $E[Y_{ij}]$ is expectation of observation ij , g_i is the effect i of genotypes assumed as random ($i^{th} = 121$ genotypes), a_j is the effect j of environment assumed as random ($j^{th} = 4$ environments), ga_{ij} is the effect of $(G \times E)$ interaction assumed as random and ϵ is the effect of random errors.

Priori

Prior distribution was normal density and the variance probability density related to model terms are:

$$g_i \sim N(0, S^{[g]})$$

$$a_j \sim N(0, S^{[a]})$$

$$(ga)_{ij} \sim N(0, S^{[ga]})$$

$$\sigma \sim HalfCauchy(0, S^{[\sigma]})$$

Priori

Parameters of prior distribution was assumed as hyperparameters:

$$S^{[g]} \sim \text{HalfCauchy}(0, \phi)$$

$$S^{[a]} \sim \text{HalfCauchy}(0, \phi)$$

$$S^{[ga]} \sim \text{HalfCauchy}(0, \phi)$$

Bayesian variance components

Table 15: Common bean UFV carioca population variance components for different traits at trials from 2020 to 2023

trait	effect	var	HPD_0.025	HPD_0.975
GA	genotype	0.031	0.018	0.047
GA	environment	8.136	0.009	20.868
GA	gen.env	0.040	0.000	0.081
GA	error	0.036	0.002	0.082
GY	genotype	18314.900	332.149	42239.694
GY	environment	3808551.300	143501.463	21978738.500
GY	gen.env	32543.119	1.586	156390.836
GY	error	193951.014	69921.263	256589.277
PA	genotype	0.035	0.019	0.053
PA	environment	180.519	0.150	1020.549
PA	gen.env	0.018	0.000	0.045
PA	error	0.032	0.007	0.059

Bayesian model fitting

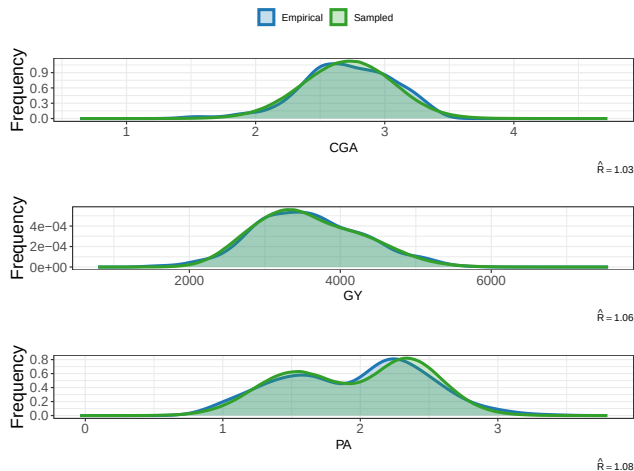


Figure 7: Bayesian model fitting results

Probability of superior performance across environments

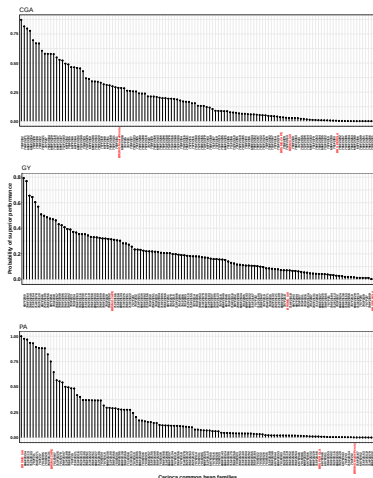


Figure 8: Probability of superior performance across environments

Probability of superior performance within environments

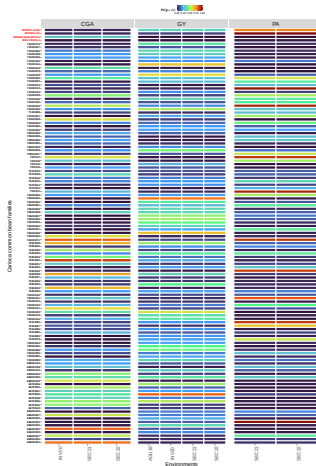


Figure 9: Probability of superior performance within environments

Pairwise probability of superior performance

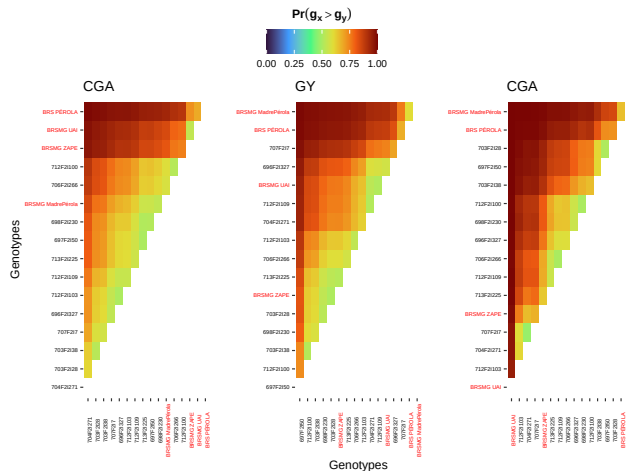


Figure 10: Pairwise probability of superior performance

Bayesian Probabilistic Selection Index

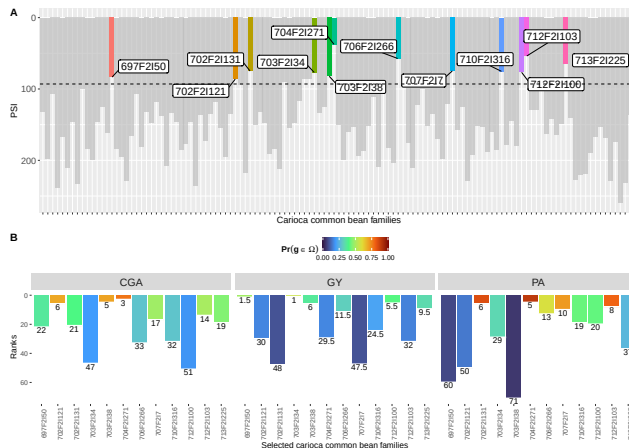


Figure 11: Probabilistic Selection Index visualization

Conclusion

Applying BPSI to carioca common bean panel, 12 families was selected aiming to extract pure lines to GA, GY and PA simultaneously. BPSI has potential to be applied on selection stage in breeding programs.

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